

DEEPESH SONAR

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Third-year Computer Engineering undergraduate working on retrieval and long-term memory for large language models. Sole author of a local-first LLM memory system and of **LSREP**, the longitudinal protocol built to evaluate it — released open-source with a public single-author arXiv preprint. Work spans constraint solvers, async services and schema design, the statistics that decide whether a change helped, and the pitch — led for six-person teams at two national and state-level competitions.

EDUCATION

Thakur College of Engineering and Technology (TCET), Mumbai

Expected 2028

B.E. in Computer Engineering | CGPA: 9.53 / 10

RESEARCH & PUBLICATIONS

LSREP: A Longitudinal State-Replay Protocol for Evaluating Conversational Memory, with ICE v2 as an Audited Local-First Architecture

Jun – Aug 2026

Sole author | arXiv preprint; LongMemEval reported

- Designed **LSREP**, a protocol that replays long-running conversations turn by turn, reconstructs memory at successive checkpoints, and scores answers against ground truth that evolves as facts are superseded; the controlled paper experiment covered **1,985 turns, 219 distinct probes, 1,211 observations, and 52 checkpoints**.
- In that paper-run comparison, ICE tied a vector-RAG baseline on answer quality (paired $\Delta = +0.00$, 95% CI $[-0.07, +0.07]$) while injecting **32% fewer fragments**; it won **30.6% vs. 21.2%** of blind comparisons.
- A density stress test found the unbudgeted baseline failed on **94.2% of 154 probes**, while ICE achieved a mean score of **4.33**; an ablation isolated rank fusion as corrective (-0.74 without fusion, recovered by $+0.82$ with fusion).
- In a matched LongMemEval diagnostic, ICE v2 scored **50.8% vs. 72.8%** for pure vector-RAG in the evidence-only oracle and **43.0% vs. 69.5%** in full-S; the finalized paper reports these as failure and quality-cost findings, not as a superiority claim.
- Paper, source, evaluation harness, and fidelity audit are public under Apache-2.0 at github.com/Deepnar/ice; preprint available on arXiv:2609.16730.

“Pixel over Paper: The Transition to Digital Note-Taking in Education and the Workplace” — top 25 of submissions selected for live presentation at TCET’s Multicon 2024 symposium; led a 4-member team through research, analysis, and delivery.

Merged open-source contributions

- ModelDock PRs #221/#222**: added Jan capability mapping and GPT4All local-model discovery with deterministic cross-platform ordering and focused tests.
- vLLM Semantic Router PR #3288**: changed Envoy’s default logging from debug to info while preserving explicit overrides; added tests and documentation.
- MNE-Python PR #14283**: fixed pytest 10 compatibility warnings in sklearn estimator checks while preserving optional-dependency behavior.
- CivicResolve PR #1** (team project; post-competition contribution): contributed to the AI Observation Engine, background worker, tests, security guards, and documentation; merged upstream after maintainer review.

PROJECTS

ICE — Infinite Context Engine

Jun 2026 – Present

Python, PyTorch, FastAPI, PostgreSQL + pgvector, SQLAlchemy 2.0, Alembic, Docker, Ollama | Apache-2.0, sole author |

github.com/Deepnar/ice | software archive: doi.org/10.5281/zenodo.21759702

- Built a local-first memory layer for LLMs — a drop-in proxy for any OpenAI-compatible client giving a locally served model persistent, structured memory across sessions, on the user’s own hardware.
- Built a **six-leg hybrid retrieval orchestrator** (BM25, decay-weighted vector, knowledge-graph traversal, procedural, document, timeline) running in parallel and fused with weighted **Reciprocal Rank Fusion**, then curated with session diversification, deduplication, and a per-query token budget.
- Trained a **multi-head PyTorch classifier** over a frozen Qwen3-Embedding-0.6B encoder: 27 sigmoid logits across 11 topic, 12 intent, and 4 context-reliance labels, feeding a calibrated gate that decides whether long-term retrieval fires at all.
- Implemented a **bi-temporal knowledge graph** — typed edges carrying valid_from/valid_until over an append-only event log — so superseded facts remain retrievable as history, supporting as-of, range and evolution queries.
- Added a per-query token budget at the assembly stage, so retrieval degrades gracefully under context pressure rather than overflowing the model’s window.

Timetable Generator — Constraint-Driven Scheduling Service

Mar 2026 – Present

FastAPI, SQLAlchemy 2.0, PostgreSQL 15, Alembic, Google OR-Tools, Pydantic, JWT, Docker | github.com/Deepnar/timetable-generator

- Designed and built a scheduling service end to end — 10+ table PostgreSQL schema, JWT auth, audit trail, publishing workflow — with one engine covering seven timetable types over a shared resource pool.
- Implemented two interchangeable solvers: a deterministic most-constrained-first greedy solver and **Google OR-Tools CP-SAT** under a 5-second class-level timeout — behind a data-driven hard-constraint registry that fails closed, plus weighted soft-constraint scoring that ranks candidate timetables.
- Guaranteed cross-timetable safety via per-resource reservation sets — faculty, room, and group keyed by (id, day, slot) — loaded from every published instance; splitting per resource is what makes “same teacher, different room” a real conflict rather than a missed one.

Presentation Forge — Academic Presentation & Report Platform

Aug 2026

Node.js, React/Vite, pptxgenjs, Ollama, SQLite, Docker | MIT | github.com/Deepnar/presentation-forge

- Built a local-first browser app and headless CLI for research-grounded academic decks and reports, with outline approval, schema validation, deterministic PPTX rendering, rasterized preview, post-generation critique, **74 semantic slide types, 34 bundled themes**, DOCX reports, local Ollama/BYOK routing, and automated renderer/API/authentication checks through GitHub Actions CI.

Prompt Routing Classifier

Mar 2026

Python, sentence-transformers, scikit-learn | github.com/Deepnar/prompt-routing-classifier

- Built an end-to-end multi-label NLP classifier routing prompts to specialised models — dataset construction, LLM-assisted labelling, training and CPU-only inference — over a ~5,100-prompt corpus.
- MiniLM-L6-v2 embeddings (384-d) into one-vs-rest logistic regression; F1 0.68 (topic) and 0.56 (intent) on a held-out 20% split. Became the basis for ICE’s production classification engine.

COMPETITIONS & LEADERSHIP

Smart India Hackathon 2025 — Team of 6

Aug – Dec 2025

- Cleared two institute-level rounds of the national hackathon; positioned the team’s civic-reporting platform against **CPGRAMS**, authored technical documentation, and led the pitch at each round.

DIPEX 2026 — Team of 6

Dec 2025 – Mar 2026

- Cleared the MMR regional round and exhibited the working model at the **DIPEX 2026 state final**; authored the platform’s **security threat model** covering location verification, recycled-image detection, AI triage, bypass vectors, and false-positive modes.
- Built the **infrastructure cost model** across report volume, AI processing, and storage scenarios; grew the team from 2 to 6 and led the pitch.

Research Cell, TCET — Technical Team

- Contributed to physical TCP/IP/networking models and helped organize two student hackathons.

Computer Society of India (CSI), TCET Chapter — Creative Committee

Jun 2025 – May 2026

- Recorded 119 officially allocated hours across chapter activities; ran social media and visual identity, including ID cards, reveal campaigns, event collateral, and logistics.

NEP Sarthi, Computer Engineering Department (Feb 2026 – present) | **Editorial team**, departmental semester magazine (two editions) | **Member**, Indian Society for Technical Education (ISTE)

EXPERIENCE

Internship Trainee — CyberPeace Foundation

Jan 2025 | Remote

- Completed a practical programme in cybersecurity awareness, OSINT technique, and cyber law and policy; hands-on training in threat identification and digital footprint analysis.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL, LaTeX; Rust (learning)

AI / ML / NLP: PyTorch, scikit-learn, sentence-transformers, NumPy, Pandas, LLMs, RAG, semantic search, embeddings, knowledge graphs, BM25, Reciprocal Rank Fusion, model evaluation, ablation design

Backend: FastAPI, SQLAlchemy 2.0, Alembic, Pydantic, JWT auth, REST API design, async Python, pytest

Databases: PostgreSQL, pgvector, MySQL, MongoDB, SQLite

Infrastructure: Docker, Docker Compose, Git, uv, Ollama, Model Context Protocol, GitHub Actions CI, Linux (Arch/CachyOS — EFI & GRUB partitioning, scripted provisioning)

Algorithms & optimization: Google OR-Tools (CP-SAT) and constraint modeling; implemented from scratch — Dijkstra, Kruskal, Floyd-Warshall, LCS, N-Queens, A* search, KMP, Rabin-Karp (github.com/Deepnar/DS-Practice)

Systems & networking: deadlock avoidance (Banker’s), page-replacement policies, memory segmentation; error detection and correction at the data-link layer (CRC, Hamming codes) with TCP and UDP socket programming

Practices: event sourcing, bi-temporal modeling, migrations, bootstrap CIs, technical writing